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1. a)

$$\frac{-1}{2}.(2+\beta+3+\alpha^2) = 1+14-6+2$$

$$\frac{-\beta-\alpha^2-5}{2} = 11$$

$$\beta+\alpha^2 = 27$$

2.) a) Volume is in three dimensional space. It is made from 3 axis x, y, z. To get volume from "k" dimension, it needs to be minimum 3 dimensional because "x=5y" creates volume on 3 dimensional space.

b) $x+y+z=6$

c) $x+4y-z-6 = x-5y$

$$9y-z=6$$

No, it doesn't create vector space.
Intersection has 3 dimension in 3 dimension.

3. a) $v = [1\ 0\ 0\ 0\ 0\ \ldots]_{1\times n}$

$$v.v^T = 1$$

$$H = 1-2.1$$

$$H = 1-2 = -1 \qquad H^2 = (-1)^2 = 1$$

b) Since $H=-1$, I is linear independent so it is orthonormal matrix.

c) It is "1" because identity matrix is made of unit vectors. For this equation. "$A.v = \lambda.v$" since eigenvector is unit vector. It is safe to say eigenvalue is 1.

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d.) It is met every condition in "$A = PDP^{-1}$".
It is linear independant, It is eigen values are 1.
Eigen vectors same as itself. It is safe to say it can be diagonalizable.

4.) a) $\frac{d}{dt}\begin{bmatrix} v \\ w \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} v \\ w \end{bmatrix}$ because it gives us much more correct results and it considers the diffusion speed.

b) $v - w = -30$

w loses %30 from itself. w lost = 12
v loses %20 too, v lost = 3
it is $\frac{7}{19}$ v, $\frac{12}{19}$ w and that equals to 63% w and 37% v

c)

"On my honer, I have neither given nor received any unauthorized and/or inappropriate assistance for all sessions of this exam. The work done on this exam is totally my own. I understand that by the school code, violation of these principles will lead to a zero grade and is subject to harsh discipline issues"

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